

EECS 489

Computer Networks

Winter 2025

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Material with thanks to Aditya Akella, Sugih Jamin, Philip Levis, Sylvia Ratnasamy, Peter Steenkiste, and many other colleagues.

Agenda

- IP routers
- Router-assisted congestion control

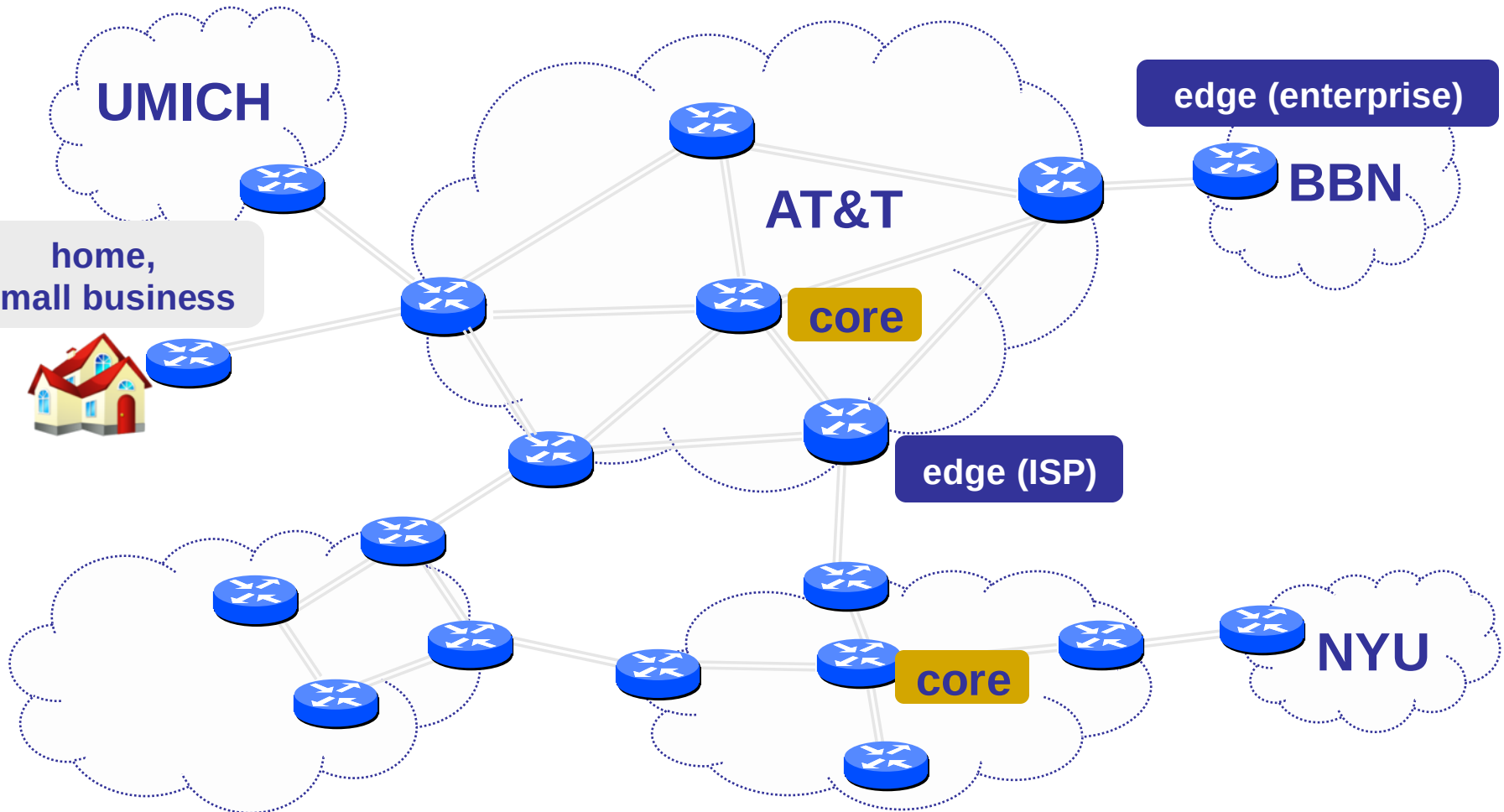
IP routers

- ▣ Core building block of the Internet infrastructure
- ▣ \$120B+ industry
- ▣ Vendors: Cisco, Huawei, Juniper (account for >90%)

Router definitions

- Router capacity = $N \times R$
- N = Number of external router “ports”
- R = Speed (“line rate”) of a port

Networks and routers



Many types of routers

□ Core

- $R = 10/40/100/400$ Gbps
- $NR = O(100)$ Tbps (Aggregated)

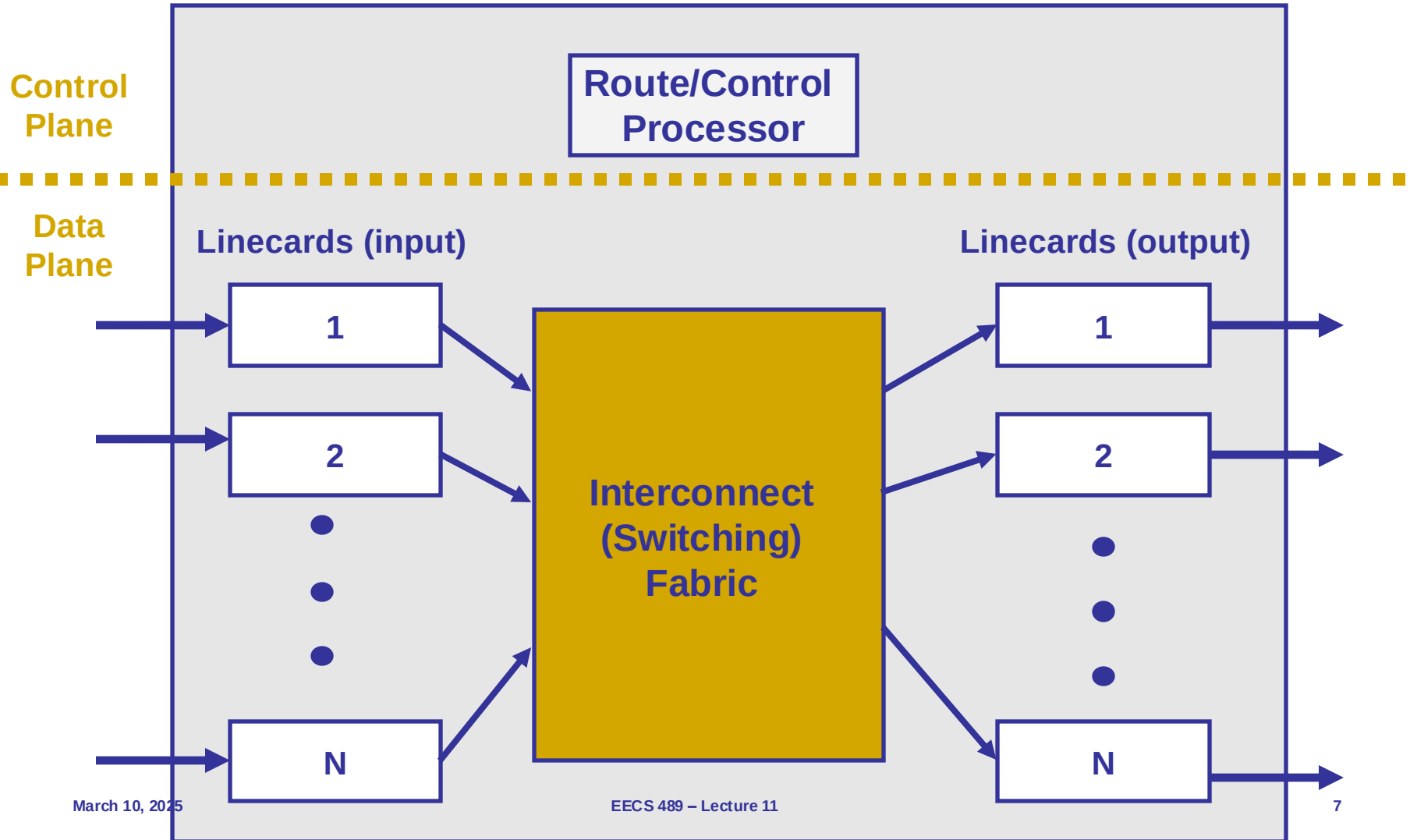
□ Edge

- $R = 1/10/40/100$ Gbps
- $NR = O(1)$ Tbps

□ Small business

- $R = 10/100/1000$ Mbps
- $NR < 10$ Gbps

What's inside a router?



What's inside a router?

□ Linecards

- Input linecards process packets on their way in
- Output linecards process packets on way out
- Input and output for the same port are on the same physical linecard

□ Interconnect/switching fabric

- Transfers packets from input to output ports

Input linecards

□ Tasks

- Receive incoming packets (physical layer stuff)
- Update the IP header
 - »TTL, Checksum, Options and Fragment (maybe)
- Lookup the output port for the destination IP address
- Queue the packet at the switch fabric

□ Challenge: speed!

- 100B packets @ 200Gbps → new packet every 4 nano secs!
- Typically implemented with specialized ASICs (network processors)

Looking up the output port

- ▣ One entry for each address → 4 billion entries!
- ▣ For scalability, addresses are aggregated

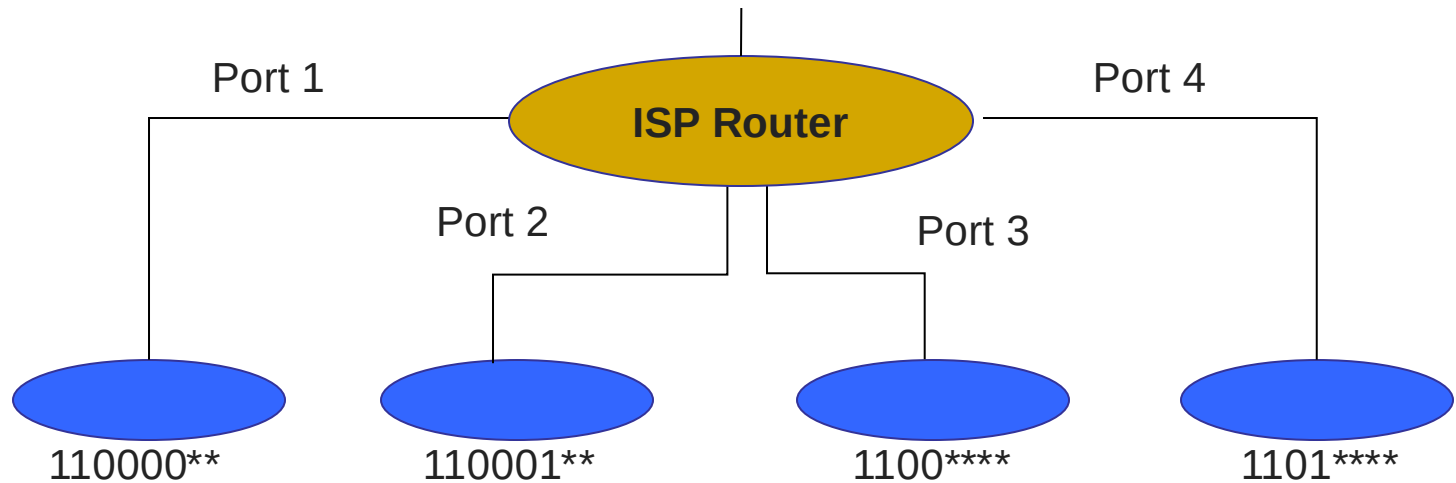
Example

- Router with 4 ports
- Destination address range mapping
 - 11 00 00 00 to 11 00 00 11: Port 1
 - 11 00 01 00 to 11 00 01 11: Port 2
 - 11 00 10 00 to 11 00 11 11: Port 3
 - 11 01 00 00 to 11 01 11 11: Port 4

Example

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- ❑ Destination address range mapping
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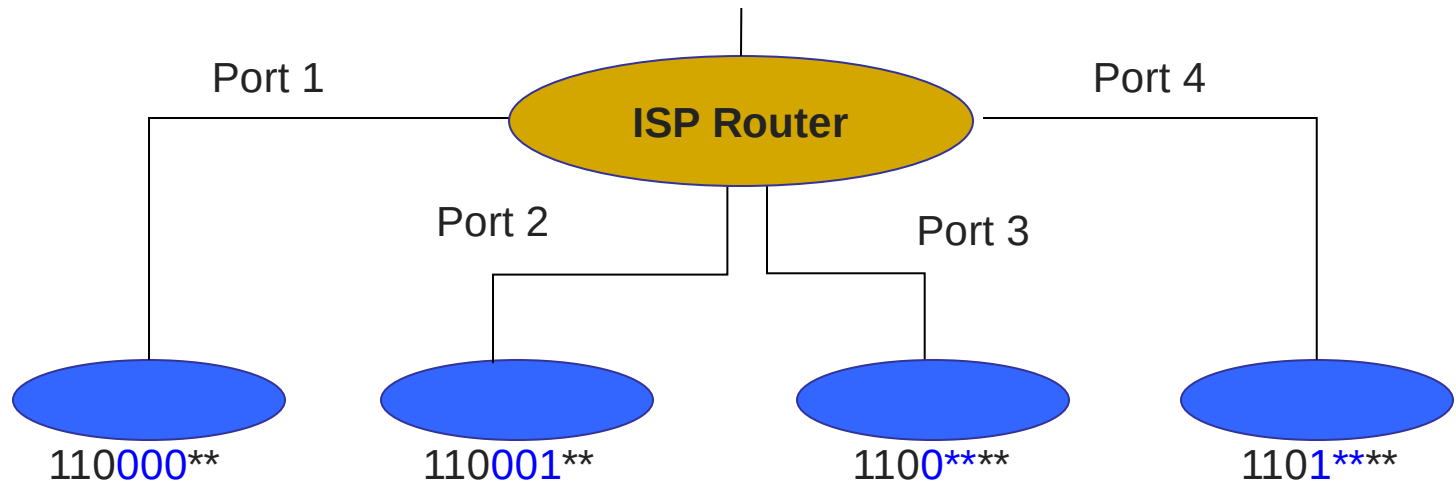
Longest prefix matching



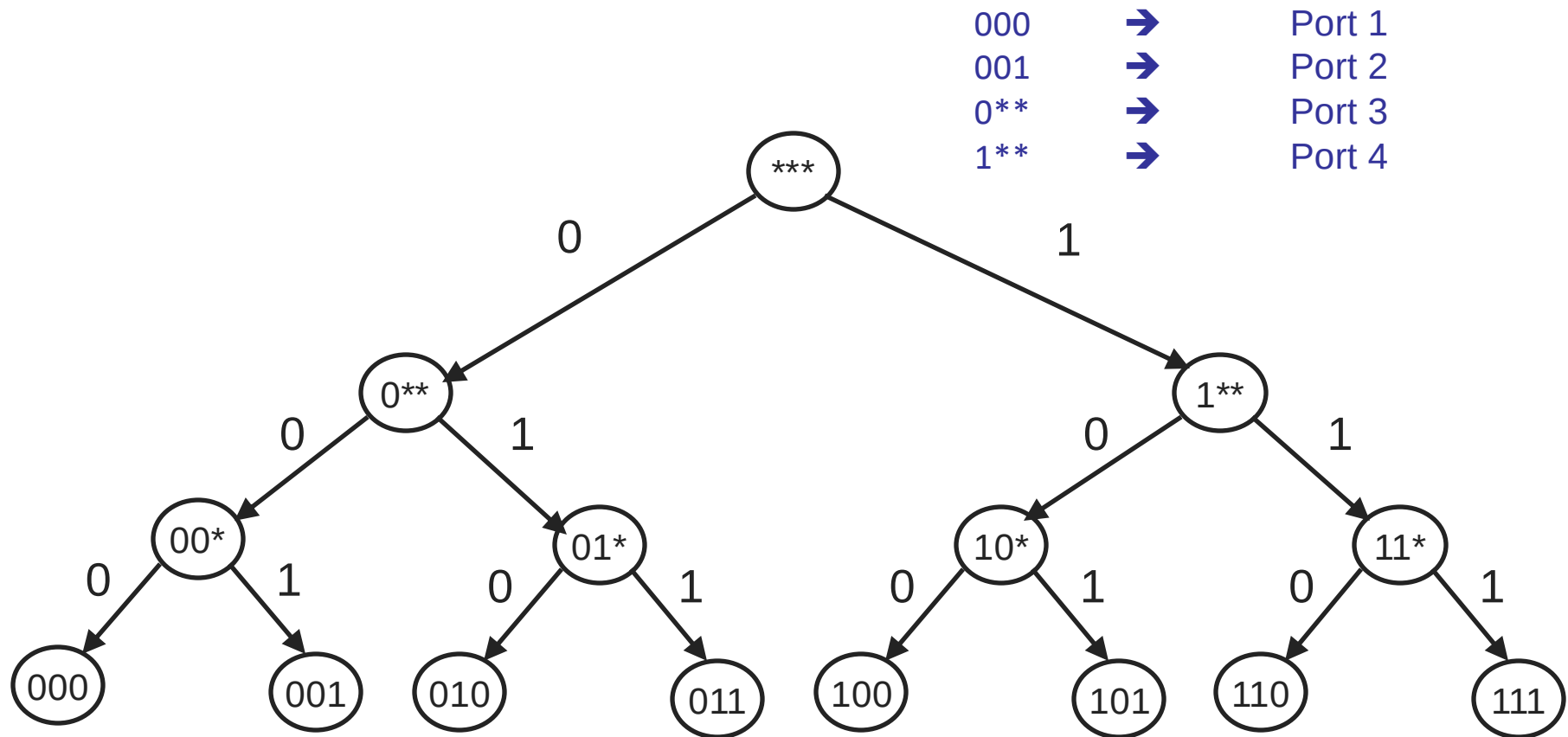
Finding match efficiently

- ❑ Testing each entry to find a match scales poorly
 - On average: $O(\text{number of entries})$
- ❑ Leverage tree structure of binary strings
 - Set up tree-like data structure

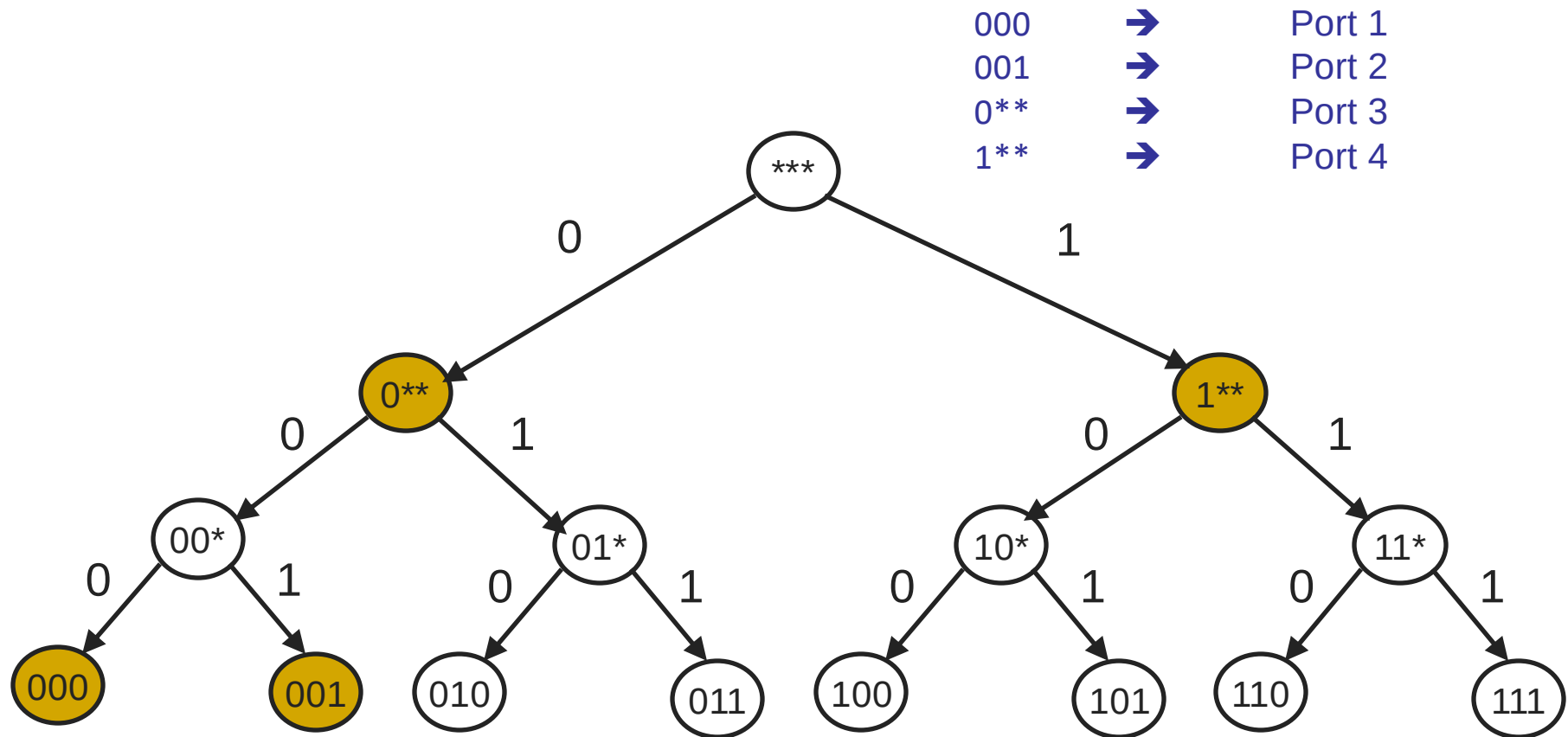
Longest prefix matching



Tree structure



Tree structure



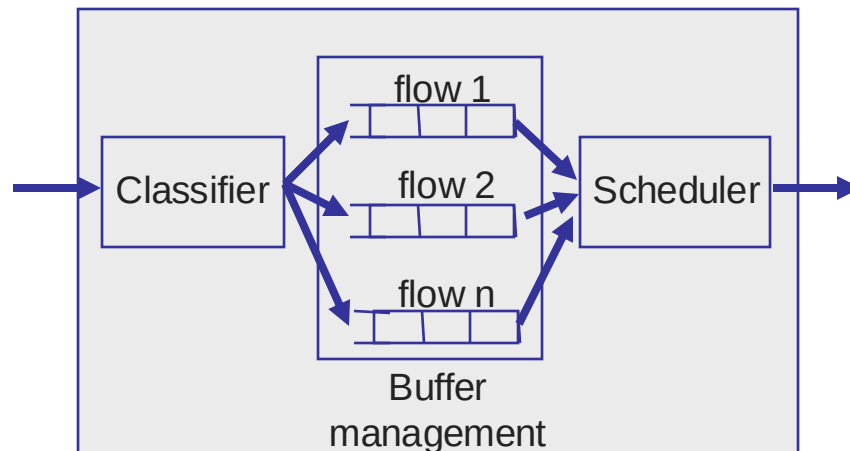
Record port associated with latest match, and only override when it matches another prefix during walk down tree

Input linecards

- Main challenge is processing speeds
- Tasks involved:
 - Update packet header (easy)
 - LPM lookup on destination address (harder)
- Mostly implemented with specialized hardware

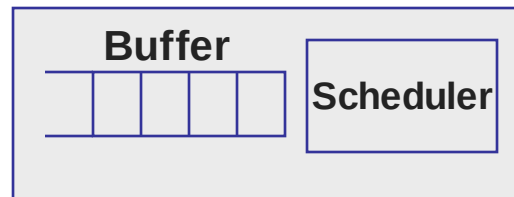
Output linecards

- ❑ **Packet classification**: map packets to flows
- ❑ **Buffer management**: decide when and which packet to drop
- ❑ **Scheduler**: decide when and which packet to transmit



Simplest: FIFO router

- ❑ No classification
- ❑ **Drop-tail buffer management**: when buffer is full drop the incoming packet
- ❑ **First-In-First-Out (FIFO) Scheduling**: schedule packets in the same order they arrive



Packet classification

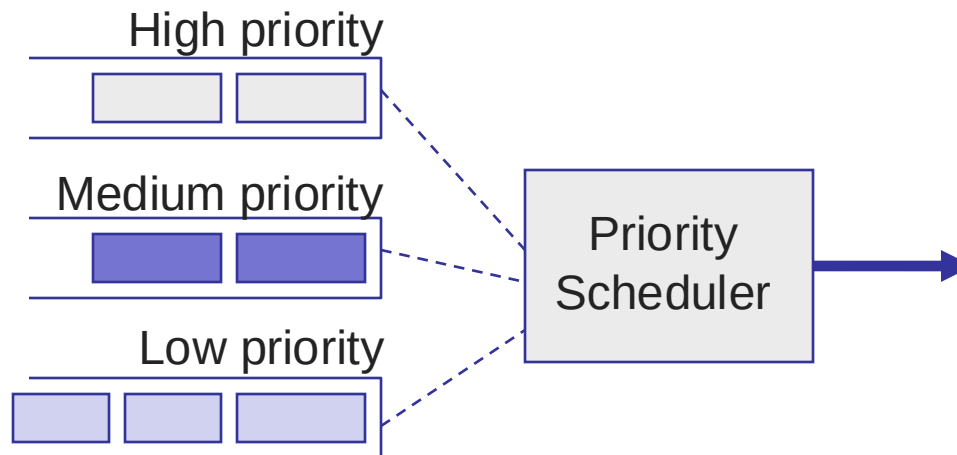
- Classify an IP packet based on a number of fields in the packet header, e.g.,
 - Source/destination IP address (32 bits)
 - Source/destination TCP port number (16 bits)
 - Type of service (TOS) byte (8 bits)
 - Type of protocol (8 bits)
- In general fields are specified by range
 - Classification requires a multi-dimensional range search!

Scheduler

- ❑ One queue per “flow”
- ❑ Scheduler decides when and from which queue to send a packet
- ❑ Goals of a scheduling algorithm
 - Fast!
 - Depends on the policy being implemented (fairness, priority, etc.)

Priority scheduler

- Priority scheduler: packets in the highest priority queue are always served before the packets in lower priority queues



Round-robin scheduler

- Round robin: packets are served from each queue in turn
- **Fair queuing (FQ)**: round-robin for packets of different size
- Weighted fair queueing (WFQ): serve proportional to weight
 - FQ gives equal weight to each flow

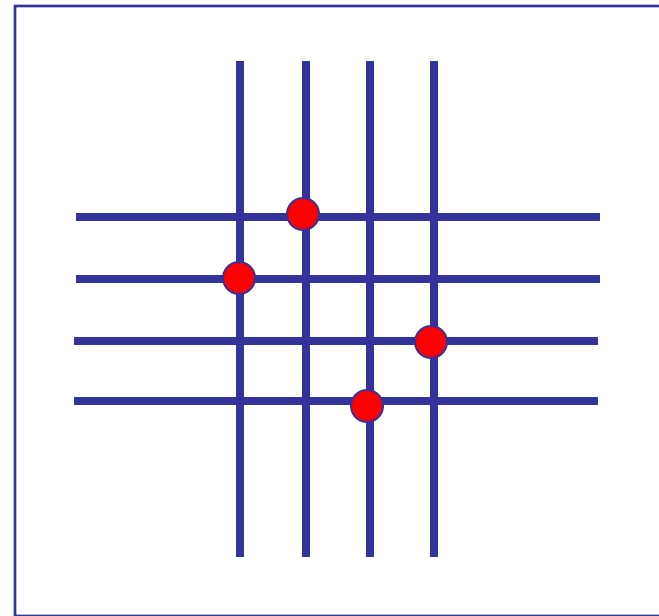
Connecting inputs to outputs: Switching fabric

- Mini-network
- Three primary ways to switch
 - Switching via shared memory
 - Switching via a bus
 - Switching via an inter-connection network
 - » For example, cross-bar

Crossbar fabric

- 2N buses intersecting with each other:
 - N input
 - N output
- Non-blocking

Input
ports



Output ports

5-MINUTE BREAK!

Announcements

- A3 is out today

ROUTER-ASSISTED CONGESTION CONTROL

Recap: TCP problems

- ❑ Misled by non-congestion losses
- ❑ Fills up queues leading to high delays
- ❑ Short flows complete before discovering available capacity
- ❑ AIMD impractical for high speed links
- ❑ Saw tooth discovery too choppy for some apps
- ❑ Unfair under heterogeneous RTTs
- ❑ Tight coupling with reliability mechanisms
- ❑ End hosts can cheat

Routers tell endpoints if they're congested

Routers tell endpoints what rate to send at

Routers enforce fair sharing

Could fix many of these with some help from routers!

Router-assisted congestion control

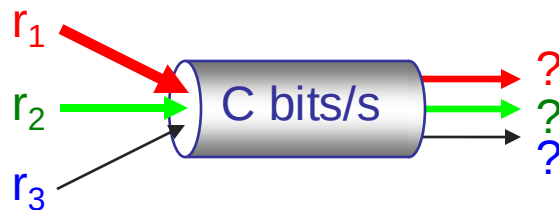
- Three tasks for congestion control
 - Isolation/fairness
 - Adjustment
 - Detecting congestion

Fairness: General approach

- | Routers classify packets into “flows”
 - Let's assume flows are TCP connections
- | Each flow has its own FIFO queue in router
- | Router services flows in a fair fashion
 - When line becomes free, take packet from next flow in a fair order
- | What does “fair” mean exactly?

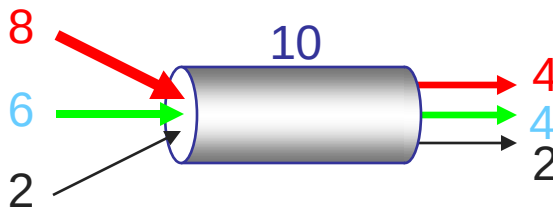
Max-Min fairness

- Given set of bandwidth demands r_i and total bandwidth C , max-min bandwidth allocations are:
 - $a_i = \min(f, r_i)$
 - where f is the unique value such that $\text{Sum}(a_i) = C$



Example

- | $C = 10$; $r_1 = 8$, $r_2 = 6$, $r_3 = 2$; $N = 3$
- | $C/3 = 3.33 \rightarrow$
 - r_3 needs only 2
 - » Can service all of r_3
 - Remove r_3 from the accounting: $C = C - r_3 = 8$; $N = 2$
- | $C/2 = 4 \rightarrow$
 - Can't service all of r_1 or r_2
 - So hold them to the remaining fair share: $f = 4$



$f = 4$:
 $\min(8, 4) = 4$
 $\min(6, 4) = 4$
 $\min(2, 4) = 2$

Max-Min fairness

- | Given set of bandwidth demands r_i and total bandwidth C , max-min bandwidth allocations are:
 - $a_i = \min(f, r_i)$
 - where f is the unique value such that $\text{Sum}(a_i) = C$
- | If you don't get full demand, no one gets more than you
- | This is what round-robin service gives if all packets are the same size

How do we deal with packets of different sizes?

- | Mental model: Bit-by-bit round robin (“fluid flow”)
- | Can you do this in practice?
 - No, packets cannot be preempted
- | But we can approximate it
 - This is what “fair queuing” routers do

Fair Queuing (FQ)

- | For each packet, compute the time at which the last bit of a packet would have left the router if flows are served bit-by-bit
- | Then serve packets in the increasing order of their deadlines

Example



Fair Queuing (FQ)

- | Implementation of round-robin generalized to the case where not all packets are equal sized
- | **Weighted fair queuing (WFQ)**: assign different flows different shares
- | Today, some form of WFQ implemented in almost all routers
 - Not the case in the 1980-90s, when CC was being developed
 - Mostly used to isolate traffic at larger granularities (e.g., per-prefix)

FQ vs. FIFO

- | FQ advantages:

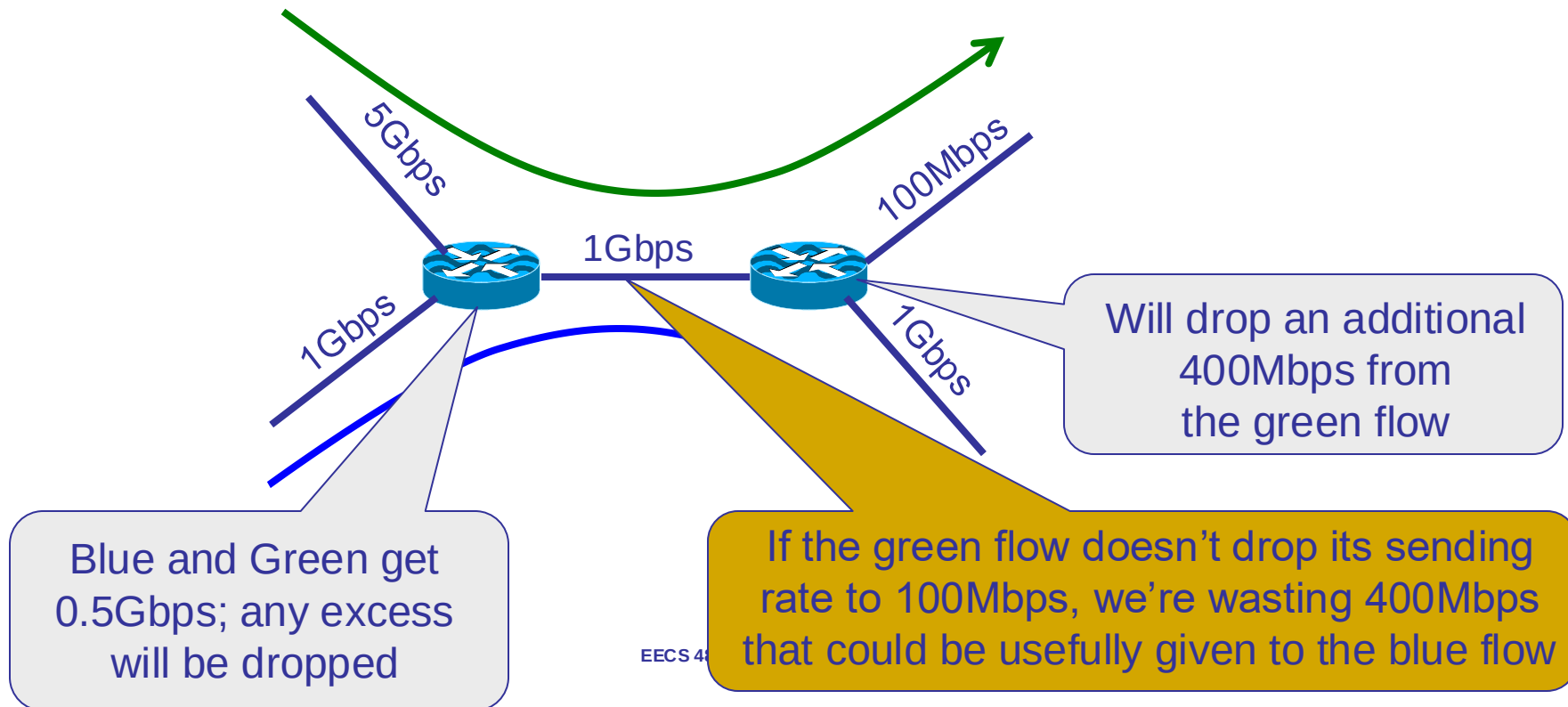
- Isolation: cheating flows don't benefit
- Bandwidth share does not depend on RTT
- Flows can pick any rate adjustment scheme they want

- | Disadvantages:

- More complex than FIFO: per flow queue/state, additional per-packet book-keeping

FQ in the big picture

- FQ does not eliminate congestion → it just manages the congestion



FQ in the big picture

- | FQ does not eliminate congestion → it just manages the congestion
 - Robust to cheating, variations in RTT, details of delay, reordering, retransmission, etc.
- | But congestion (and packet drops) still occurs
- | We still want end-hosts to discover/adapt to their fair share!

Fairness is a controversial goal

- | What if you have 8 flows, and I have 4?
 - Why should you get twice the bandwidth?
- | What if your flow goes over 4 congested hops, and mine only goes over 1?
 - Why shouldn't you be penalized for using more scarce bandwidth?
- | What is a flow anyway?
 - TCP connection
 - Source-Destination pair?
 - Source?

Router-Assisted Congestion Control

- | CC has three different tasks:
 - Isolation/fairness
 - Rate adjustment
 - Detecting congestion

Why not let routers tell what rate end hosts should use?

- | Packets carry “rate field”
- | Routers insert “fair share” f in packet header
- | End-hosts set sending rate (or window size) to f
 - Hopefully (still need some policing of end hosts!)
- | This is the basic idea behind the “Rate Control Protocol” (RCP) from Dukkkipati et al. '07
 - Flows react faster

Router-Assisted Congestion Control

- | CC has three different tasks:
 - Isolation/fairness
 - Rate adjustment
 - Detecting congestion

Explicit Congestion Notification (ECN)

- | Single bit in packet header; set by congested routers
 - If data packet has bit set, then ACK has ECN bit set
- | Many options for when routers set the bit
 - Tradeoff between (link) utilization and (packet) delay
- | Congestion semantics can be exactly like that of drop
 - i.e., end-host reacts as though it saw a drop

ECN

| Advantages:

- Don't confuse corruption with congestion; recovery w/ rate adjustment
- Can serve as an early indicator of congestion to avoid delays
- Easy (easier) to incrementally deploy
 - » Today: defined in RFC 3168 using ToS/DSCP bits in the IP header
 - » Common in datacenters

Summary

- | IP routers form the backbone of the Internet
- | Aims for speed while providing fairness
- | Routers can assist in addressing/mitigating many of TCP's shortcomings